

An La

 [anla-cs.github.io](https://github.com/anla-cs) |  [anla11](#) |  [anla11](#) |  langocthuyan@gmail.com |  anla@umass.edu

EDUCATION

2021 - present MS/PhD (Computer Science) at **University of Massachusetts Amherst** (GPA: 3.82/4.0)
Advisor: [Dr. Hung Le](#)

2013 - 2017 Bachelor's Degree at **Honors Program, VNU-HCMUS, Vietnam** (GPA: 3.57/4.0)
Information Technology, Graduated with distinction.

PUBLICATIONS

La, A., & Le, H. *Dynamic Locality Sensitive Orderings in Doubling Metrics*. To be appeared in [STOC 2025](#).

La, A., & Le, H. (2024, August). *New weighted additive spanners*.

La, A., Vo, P., & Vu, T. (2019, July). *Adaptive Collaborative Filtering for Recommender System*. In International Conference on Conceptual Structures (pp. 117-130).

La, A. N. T.*, Nguyen, D. P.*, Pham, N. M., & Vu, Q. H. (2018). *Multi-modal video retrieval using Dilated Pyramidal Residual network*. Science and Technology Development Journal-Natural Sciences, 2(5), 138-143. ¹

SKILLS

Theory Data structures & algorithms, quantitative analysis (Bayesian statistics, causal inference), probabilistic graphical models.

Programming Proficient in Python to: develop recommendation systems; build machine learning models (Tensor Flow, PyTorch, Scikit-Learn); process, analyze and visualize data (Numpy, Pandas, Seaborn).
Proficient in C++ to implement algorithms in competitive programming contests and Image Processing project (with OpenCV).

EXPERIENCE

2021 - now: Research and Teaching Assistant at [Theory CS Group/UMass Amherst](#)

- ★ Study data structures & algorithms in computational geometry and their applications in data mining.
- ★ Study online algorithms with machine learning predictions.
- ★ Teaching Assistant: [Algorithms for Data Science \(Fall22, Spring25\)](#), [Advanced Algorithms \(Spring23\)](#).
- ★ Technical skills: *algorithm design, data structure design, algorithmic analysis*.

2020 - 2021: Data Scientist at PrimeData, Vietnam

github.com/anla11/analytic_marketing

- ★ Designed and implemented an automatic framework for segment analytics.
- ★ Generated insightful segments of users without manual analysis for algorithmic marketing applications, such as business identity, customer engagement campaign.
- ★ Technical skills: *quantitative analysis, Bayesian machine learning and probabilistic programming*.

2017 - 2019: Data Scientist at FPT Telecom, FPT Group, Vietnam

github.com/anla11/adaptive_cf_recsys

- ★ Designed and implemented a graph-based model dealing with multiple evaluation metrics for the recommender system of fptplay.vn.
- ★ Increased precision by 6% while maintaining diversity, coverage, and congestion.
- ★ Technical skills: *content-based analysis and modelling, user-centric analysis and collaborative-filtering modelling, graph-based algorithms, performance evaluation analysis*.

SELECTIVE AWARDS

Dec. 2016 National Vietnam award for Outstanding Female Students in Science and Technology

Aug. 2016 Awards from Facebook Hackathon Vietnam 2016

1st prize of Most Innovative Product

2nd prize of Best Product in Facebook Marketing Category

2014 *2nd prize* in ACM-ICPC Vietnam National 1st Round

2013 *3rd prize* in Informatics at the Vietnam National Excellent Student Exam

^{1*}These authors contributed equally to the work.